SOP : AzCopy and File Transfer to Azure Blob Storage

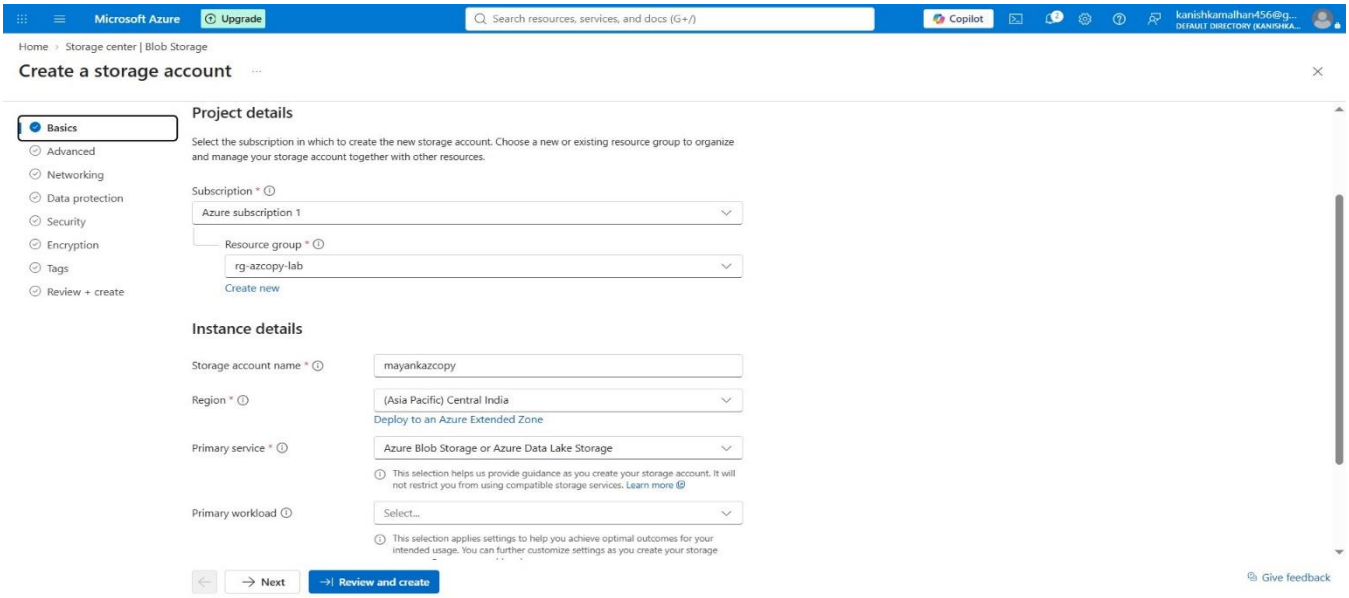
Objective

To install and configure AzCopy, upload local files and folders to Azure Blob Storage, download files from Azure to a local computer, perform recursive transfers, and verify the results.

Step-by-Step Procedure

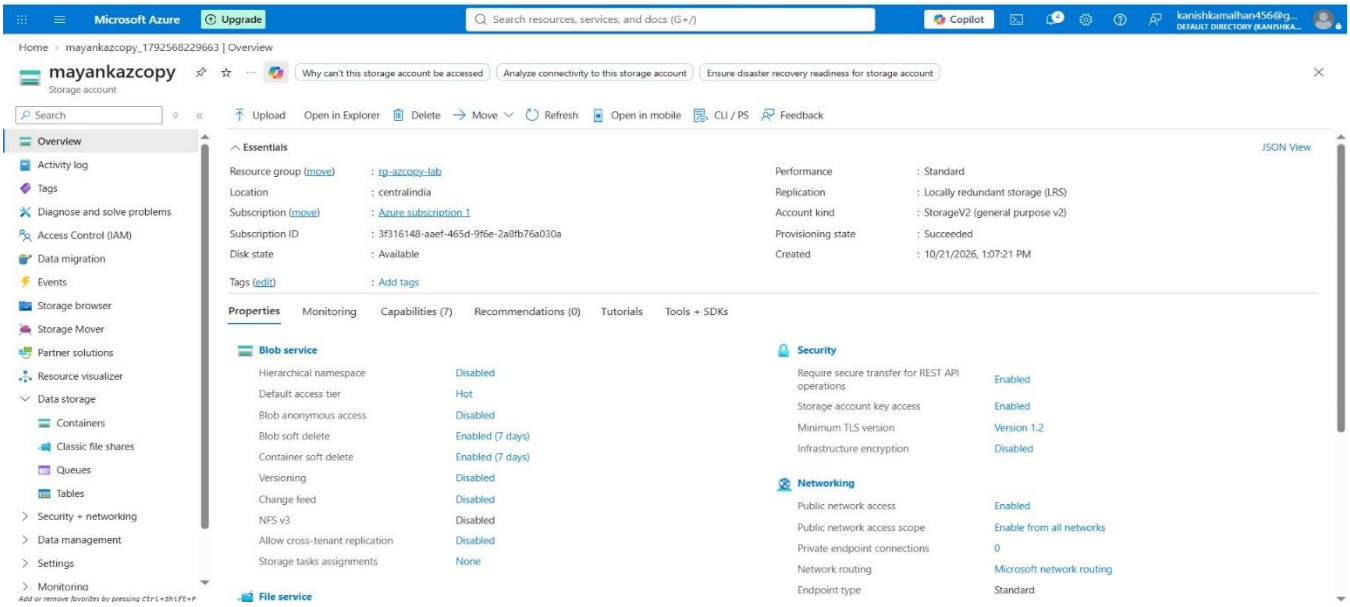
Step 1: Create a Storage Account

1. Sign in to the Azure Portal.
2. Search for Storage accounts and select Create.
3. Select the subscription and resource group.
4. Enter mayankazcopy as the storage account name.
5. Select the region and configure the required settings.
6. Click Review + create, then click Create.



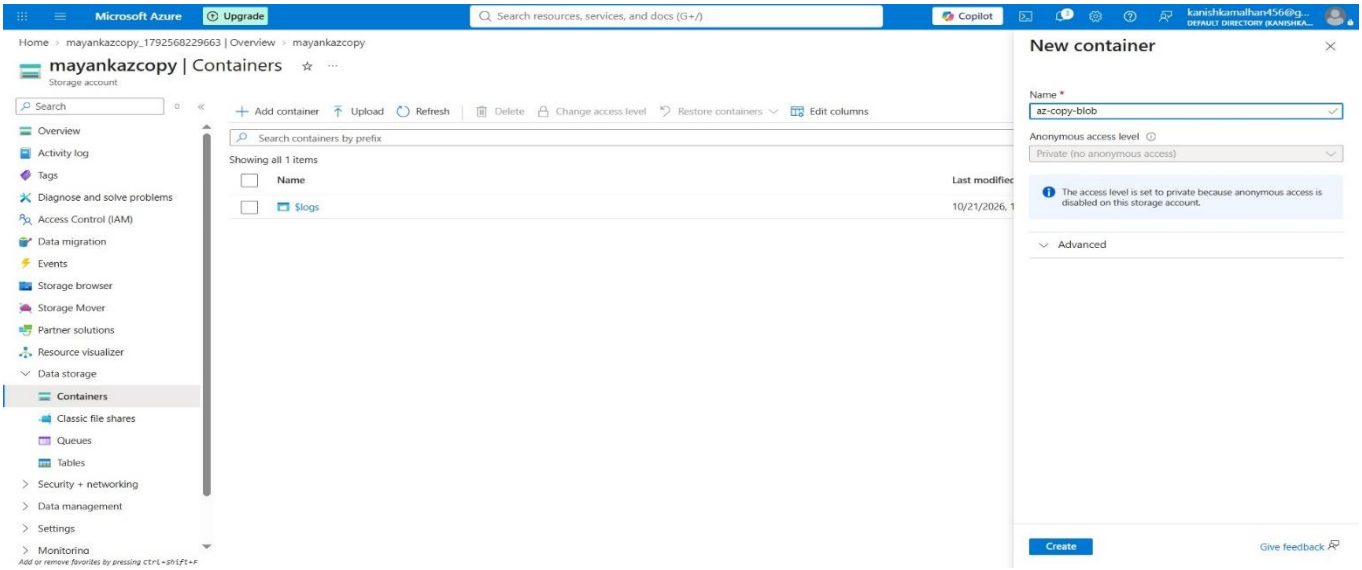
Step 2: Verify the Storage Account

1. Open the mayankazcopy storage account.
2. Go to the Overview page.
3. Verify the resource group, region, performance tier, and replication setting.
4. Confirm that the provisioning state is Succeeded.



Step 3: Create a Blob Container

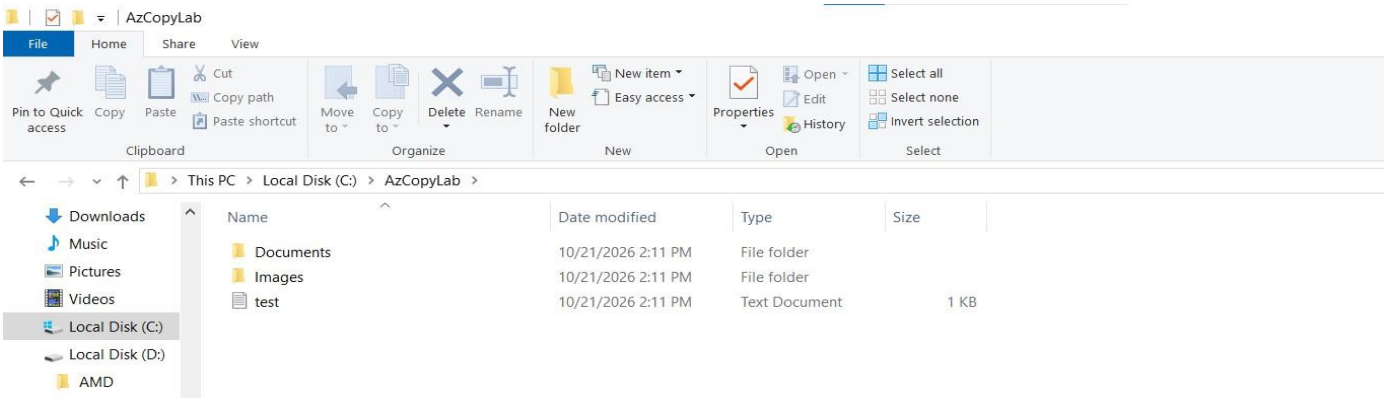
1. Open the storage account and select Data storage → Containers.
2. Click Add container.
3. Enter the container name az-copy-blob.
4. Keep the anonymous access level set to Private (no anonymous access).
5. Click Create.



Step 4: Create Local Test Files and Folders

1. Open Windows PowerShell.
2. Create a local directory named C:\AzCopyLab.
3. Create two subfolders: Documents and Images.

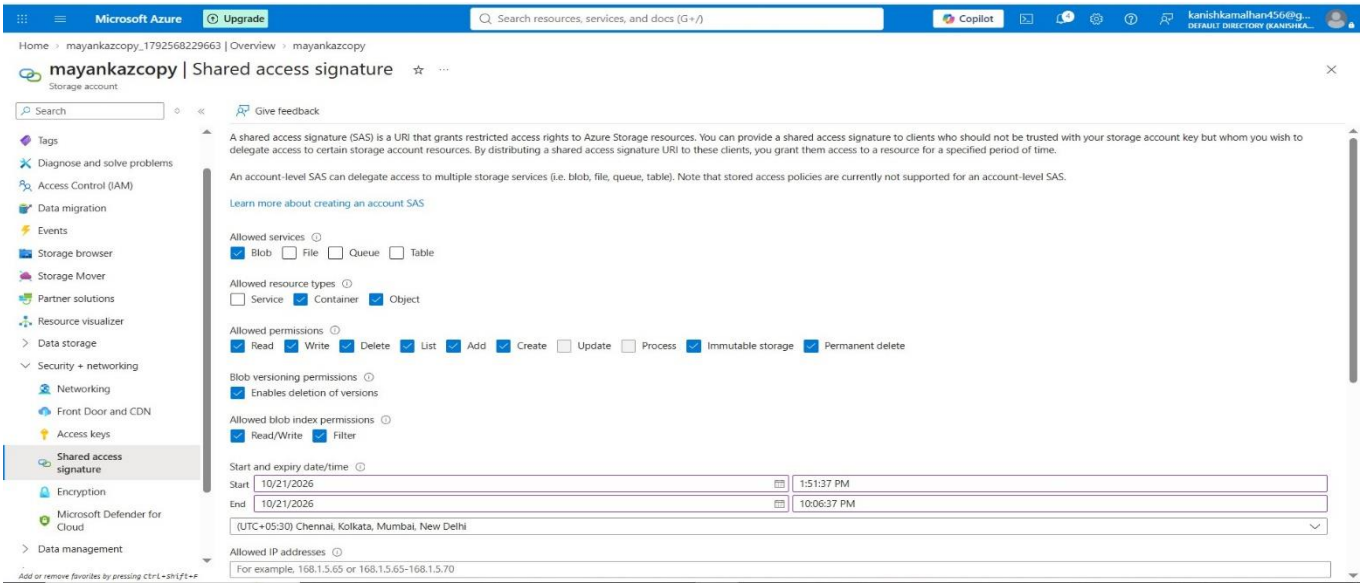
4. Create the following sample files:
   - test.txt
   - Documents\document.txt
   - Images\image-info.txt
5. Open File Explorer and verify that the files and folders have been created.



Step 5: Generate a Shared Access Signature (SAS) Token

1. Open the mayankazcopy storage account in Azure Portal.
2. Under Security + networking, select Shared access signature.
3. Select Blob as the allowed service.
4. Select Container and Object as the allowed resource types.
5. Enable the required permissions: Read, Write, Create, and List.
6. Set the start and expiry date/time.
7. Generate the SAS token and copy it securely.

Security note: Never include the full SAS token or connection string in the submitted SOP. Treat these credentials like a password.



Step 6: Upload Local Files to Azure Blob Storage

1. Open PowerShell in the folder where azcopy.exe is located.
2. Define the SAS token and Blob Container URL.

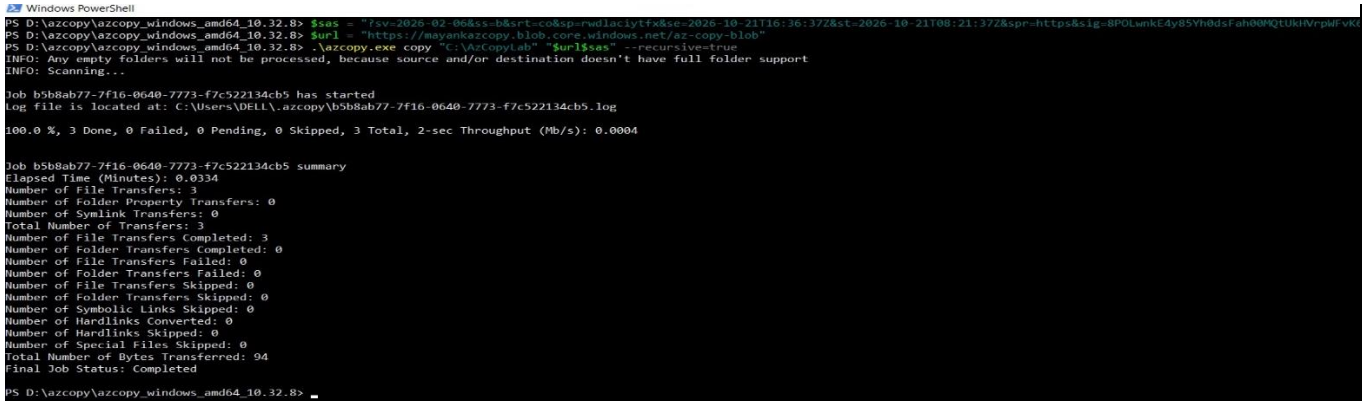
$sas = "?PASTE_YOUR_SAS_TOKEN_HERE"
$url = "https://mayankazcopy.blob.core.windows.net/az-copy-blob"

Replace the placeholder with your actual SAS token. If the copied token already starts with ?, do not add another question mark.

3. Upload the local folder and its contents recursively.

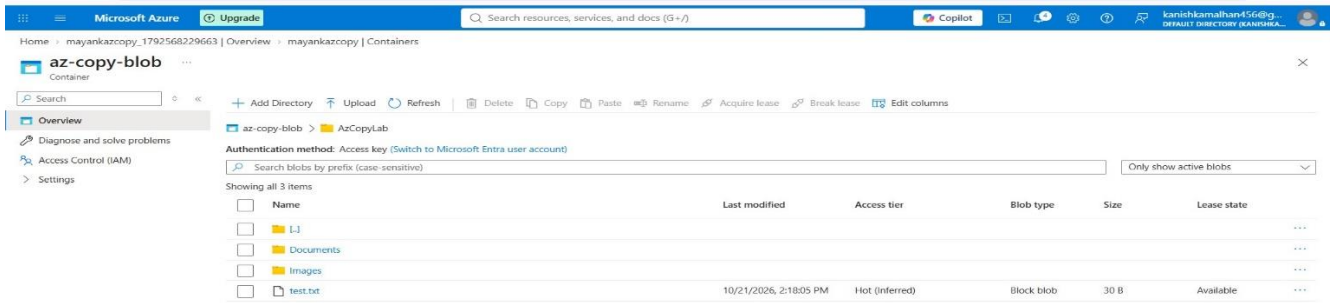
.\azcopy.exe copy "C:\AzCopyLab" "urlsas" --recursive=true

4. Wait for the transfer to finish.
5. Verify the summary shows 3 completed file transfers and 0 failed transfers.



Step 7: Verify Uploaded Files in Azure Portal

1. Open the mayankazcopy storage account.
2. Select Containers → az-copy-blob.
3. Open the AzCopyLab folder.
4. Verify that test.txt, Documents, and Images are present.
5. Check that the files inside the subfolders are also available.

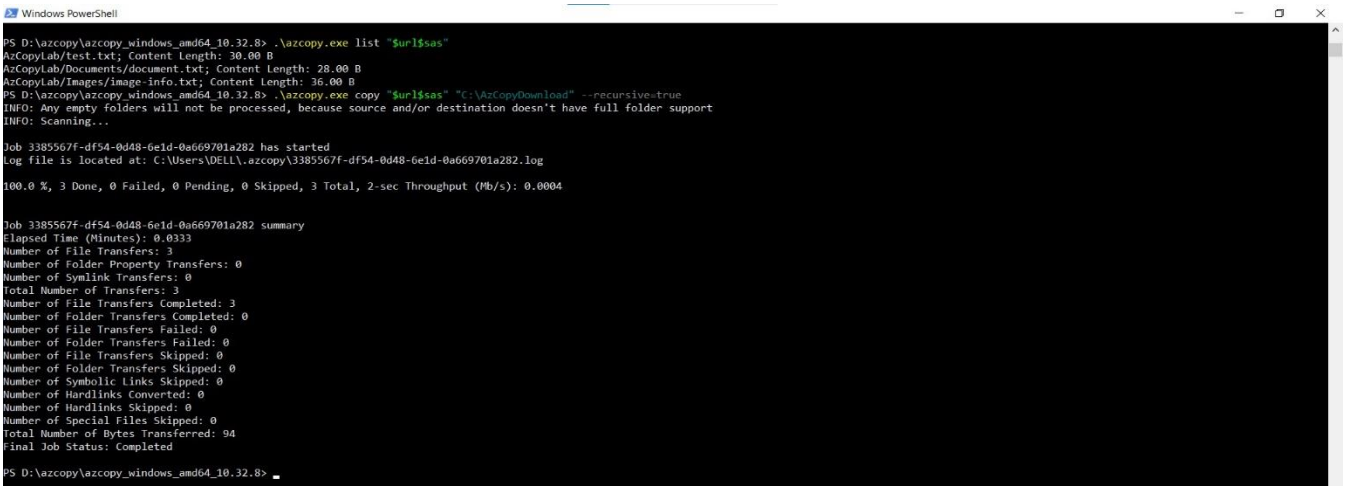


Step 8: Download Files from Azure to the Local Computer

1. Return to the same PowerShell window.
2. Run the following command to download the container contents recursively.

.\azcopy.exe copy "urlsas" "C:\AzCopyDownload" --recursive=true

3. Wait until the transfer finishes.
4. Verify that the output reports 3 completed file transfers and 0 failed transfers.



Step 9: Verify Downloaded Files and Folders Locally

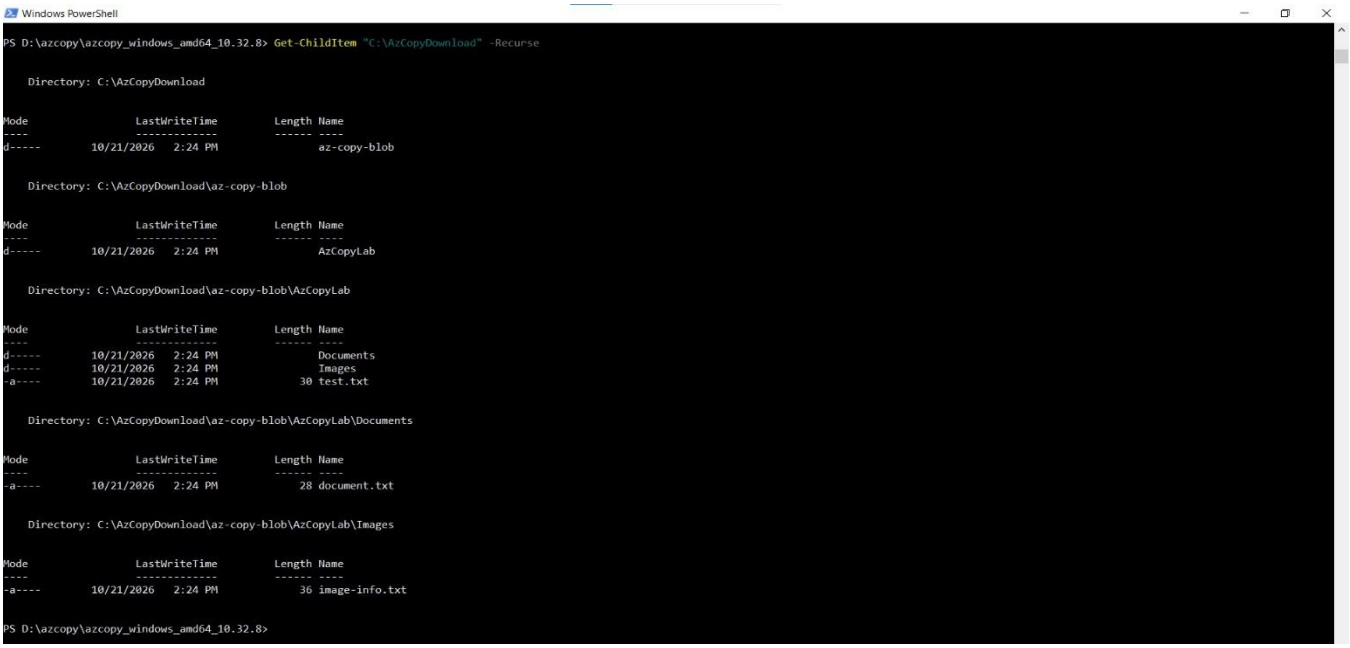
1. Run the following command in PowerShell to display all downloaded files and subfolders.

Get-ChildItem "C:\AzCopyDownload" -Recurse

2. Verify that a new az-copy-blob folder has been created inside C:\AzCopyDownload.
3. Confirm that the downloaded structure contains:

```
C:\AzCopyDownload
└── az-copy-blob
    └── AzCopyLab
        ├── test.txt
        ├── Documents
        │   └── document.txt
        └── Images
            └── image-info.txt
```

4. Confirm that all three files are present in the downloaded folders.



Result

AzCopy was installed and used to transfer files between the local Windows computer and Azure Blob Storage. Three files were uploaded recursively and downloaded successfully. The local verification confirmed that the downloaded files and folder structure were present.